

JDBC CallableStatement Assignment - Complete Solution

Problem Statement

Product Inventory Management System using JDBC CallableStatement.

Project Structure

```
ProductInventory
├──
├── ProductCallableDemo.java
├── MySQL Database (inventorydb)
```

Create Database and Table

```
create database inventorydb;

use inventorydb;

create table products
(
    product_id int,
    product_name varchar(50),
    category varchar(30),
    quantity int,
    price double
);
```

Insert Sample Data

```
insert into products values
(101, 'Wireless Mouse', 'Electronics', 50, 799),
(102, 'Keyboard', 'Electronics', 40, 999),
(103, 'Office Chair', 'Furniture', 15, 4500),
(104, 'Study Table', 'Furniture', 10, 6500),
(105, 'Water Bottle', 'Home', 100, 299),
(106, 'LED Bulb', 'Home', 75, 199),
(107, 'Cricket Bat', 'Sports', 20, 1800),
(108, 'Football', 'Sports', 25, 1200),
(109, 'Backpack', 'Accessories', 35, 999),
(110, 'Travel Bag', 'Accessories', 18, 2200);
```

Stored Procedure

```
delimiter $$

create procedure GetProductsByCategory
(
    in p_category varchar(30)
)
begin
    select *
    from products
    where category = p_category;

end $$

delimiter ;
```

Java Program - ProductCallableDemo.java

```

import java.sql.*;
import java.util.Scanner;

public class ProductCallableDemo {

    public static void main(String[] args)
    throws Exception {

        Scanner sc = new Scanner(System.in);

        Class.forName("com.mysql.cj.jdbc.Driver");

        Connection con =
        DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/inventorydb",
        "root",
        "root");

        System.out.println("Enter Category : ");

        String category = sc.nextLine();

        CallableStatement cs =
        con.prepareCall(
        "{call GetProductsByCategory(?)}");

        cs.setString(1, category);

        ResultSet rs =
        cs.executeQuery();

        while(rs.next()) {

            System.out.println(
            rs.getInt(1) + " " +
            rs.getString(2) + " " +
            rs.getString(3) + " " +
            rs.getInt(4) + " " +
            rs.getDouble(5));

        }

        con.close();

    }
}

```

Expected Output Example

Input:
Electronics

Output:
101 Wireless Mouse Electronics 50 799
102 Keyboard Electronics 40 999

Viva Questions

1. What is CallableStatement?
Used to call stored procedures from Java.
2. Syntax:

```
CallableStatement cs =
con.prepareCall("{call procedure(?)}");
```
3. Difference between PreparedStatement and CallableStatement:
 PreparedStatement -> Executes SQL queries.
 CallableStatement -> Executes Stored Procedures.
4. Why use Stored Procedure?
 Reusable database logic and better performance.